

Question 1.

$$I_{D3} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$100 \times 10^{-6} = \frac{1}{2} \times 135 \times 10^{-6} \times \frac{100}{0.5} (V_{GS} - 0.7)^2$$

$$V_{GS3} = V_X = \underline{0.786 \text{ V}} \quad \text{Ans}$$

$$I_{D2} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$0.5 \times 10^{-3} = \frac{1}{2} \times 135 \times 10^{-6} \times \frac{100}{0.5} (V_{GS} - 0.7)^2$$

$$V_{GS2} = 0.892 \text{ V}$$

$$\therefore V_{G2} = V_X + V_{GS2} = 0.786 + 0.892 = \underline{1.678 \text{ V}} \quad \text{Ans}$$

$$I_2 = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{OV2})^2$$

$$0.5 \times 10^{-3} = \frac{1}{2} \times 40 \times 10^{-6} \times \frac{50}{0.5} (V_{OV2})^2$$

$$V_{OV2} = 0.5 \text{ V}$$

$$V_{out\max} = V_{DD} - V_{OV2} = 3 - 0.5 = 2.5 \text{ V}$$

$$V_{out\min} = V_{GS3} + V_{OV2} = 0.786 \text{ V} + 0.892 - 0.7 = 0.978 \text{ V}$$

$$\text{Output voltage swing} = V_{out\max} - V_{out\min} = 2.5 \text{ V} - 0.978 \text{ V} = 1.522 \text{ V} \quad \text{Ans}$$

$$\text{Gain} = -g_{m1} [R_{up} \parallel R_{down}]$$

$$R_{up} = r_{O2}$$

$$R_{down} = g_{m3} [r_{O3} \parallel r_{O1}] g_{m2} r_{O2} r_{O1}$$

$$g_{m1} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_{D1}} = \sqrt{2 \times 135 \times 10^{-6} \times \frac{100}{0.5} \times 0.5 \times 10^{-3}} = 5.2 \text{ mS}$$

$$r_{O2} = \frac{1}{\lambda_p I_2} = \frac{1}{0.2 \times 0.5 \times 10^{-3}} = 10 \text{ K}\Omega$$

$$g_{m3} = \sqrt{2 \times 135 \times 10^{-6} \times \frac{100}{0.5} \times 100 \times 10^{-6}} = 2.32 \text{ mS}$$

$$g_{m2} = 5.2 \text{ mS}$$

$$r_{O3} = \frac{1}{\lambda_n I_{D3}} = \frac{1}{0.1 \times 100 \times 10^{-6}} = 100 \text{ K}\Omega$$

$$r_{O1} = \frac{1}{\lambda_p I_1} = \frac{1}{0.2 \times 100 \times 10^{-6}} = 50 \text{ K}\Omega$$

$$r_{O2} = r_{O1} = \frac{1}{\lambda_n I_{D2}} = \frac{1}{0.1 \times 0.5 \times 10^{-3}} = 20 \text{ K}\Omega$$

$$R_{down} = 2.32 \text{ mS} [100 \text{ K} \parallel 50 \text{ K}] 5.2 \text{ mS} \times 20 \text{ K} \times 20 \text{ K} = 160.85 \text{ M}\Omega$$

$$\therefore \text{Gain} = -5.2 \text{ mS} [10 \text{ K} \parallel 160.85 \text{ M}\Omega] = -51.99 \approx -52 \text{ V/V} \quad \text{Ans}$$

Question 2 :

$$\textcircled{1} \quad C_c \geq 0.22 C_L$$

$$C_c \geq 0.22 \times 5 \text{ pF}$$

$$C_c \geq 1.1 \text{ pF}$$

$$\underline{C_c = 1.1 \text{ pF}}$$

$$\textcircled{2} \text{ Slew rate} = \frac{I_5}{C_c}$$

$$\therefore I_5 = \text{SR} \times C_c = 50 \frac{\text{V}}{\mu\text{s}} \times 1.1 \text{ pF} = 55 \mu\text{A}$$

$$\textcircled{3} \quad GB = g_{m2}/C_c$$

$$\therefore g_{m2} = GB \times C_c = 2\pi \times 50 \times 10^6 \times 1.1 \times 10^{-12} = 0.345 \text{ mS}$$

$$g_{m2} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} \times I_D}$$

$$0.345 \times 10^{-3} = \sqrt{2 \times 300 \times 10^{-6} \times \frac{W}{L} \times 27.5 \times 10^{-6}}$$

$$\underline{\left(\frac{W}{L}\right)_{1,2} = 7.21}$$

$$\textcircled{4} \quad \text{ICMR}(+) = V_{DD} - \sqrt{\frac{2I_{D3}}{\mu_p C_{ox} (W/L)_3}} - |V_{th3, \max}| + V_{th1, \min}$$

$$\therefore \left(\frac{W}{L}\right)_{3,4} = \frac{2I_{D3}}{\mu_p C_{ox} [V_{DD} - \text{ICMR}(+) - |V_{th3, \max}| + V_{th1, \min}]^2}$$

$$= \frac{2 \times 27.5 \times 10^{-6}}{60 \times 10^{-6} [1.8 - 1.6 - 0.51 + 0.47]^2}$$

$$\underline{\underline{\left(\frac{W}{L}\right)_{3,4} = 35.81}}$$

$$\textcircled{5} \quad \text{ICMR}(-) = V_{OV5} + V_{GS1}$$

$$= V_{OV5} + \sqrt{\frac{2I_{D1}}{\mu_n C_{ox} (W/L)_1}} + V_{th1, \max}$$

$$\therefore V_{OV5} = \text{ICMR}(-) - \sqrt{\frac{2I_{D1}}{\mu_n C_{ox} (W/L)_1}} - V_{th1, \max}$$

$$= 0.8 - \sqrt{\frac{2 \times 27.5 \times 10^{-6}}{300 \times 10^{-6} \times 7.21}} - 0.59$$

$$= 0.151 \text{ V}$$

$$I_{D5} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_5 (V_{OV5})^2$$

$$55 \times 10^{-6} = \frac{1}{2} \times 300 \times 10^{-6} \times \left(\frac{W}{L}\right)_5 \times (0.151)^2$$

$$\underline{\underline{\left(\frac{W}{L}\right)_{5,6} = 16.08}}$$

$$\textcircled{6} \quad g_{m7} \geq 10 g_{m1}$$

$$\geq 10 \times 0.345 \text{ mS}$$

$$\therefore g_{m7} = 3.45 \text{ mS}$$

$$I_{D3} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_3 (V_{OV3})^2$$

$$27.5 \times 10^{-6} = \frac{1}{2} \times 60 \times 10^{-6} \times 35.81 \times (V_{OV3})^2$$

$$V_{OV3} = 0.16 \text{ V}$$

$$\therefore V_{OV7} = 0.16 \text{ V}$$

$$g_{m7} = \mu_p C_{ox} \frac{W}{L} V_{OV7}$$

$$3.45 \times 10^{-3} = 60 \times 10^{-6} \times \frac{W}{L} \times 0.16$$

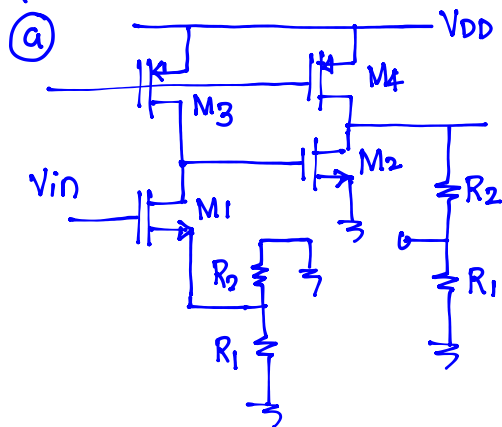
$$\left(\frac{W}{L} \right)_7 = \underline{\underline{359.38}}$$

$$\begin{aligned} I_{D7} &= \frac{1}{2} \mu_p C_{ox} \frac{W}{L} \times (V_{OV7})^2 \\ &= \frac{1}{2} \times 60 \times 10^{-6} \times 359.38 \times (0.16)^2 \\ &= 276 \text{ } \mu\text{A} \end{aligned}$$

$$\frac{I_{D8}}{I_{D6}} = \frac{276 \times 10^{-6}}{55 \times 10^{-6}} = \frac{(W/L)_8}{(W/L)_6}$$

$$\therefore \left(\frac{W}{L} \right)_8 = \underline{\underline{80.69}}$$

Question 3:



$$A = - \frac{r_{o3}}{(R_1 \parallel R_2) + \frac{1}{g_{m1}}} \times - g_{m2} [r_{o4} \parallel r_{o2} \parallel (R_1 + R_2)]$$

$$r_{on} = \frac{1}{\lambda_n I_D} = \frac{1}{0.1 \times 0.5 \times 10^{-3}} = 20 \text{ K}$$

$$r_{op} = \frac{1}{\lambda_p I_D} = \frac{1}{0.2 \times 0.5 \times 10^{-3}} = 10 \text{ K}$$

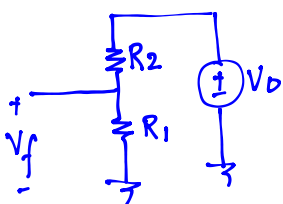
$$g_{m1} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D}$$

$$= \sqrt{2 \times 135 \times 10^{-6} \times \frac{50}{0.5} \times 0.5 \times 10^{-3}} = 3.67 \text{ mS} = g_{m2}$$

$$\therefore A = \frac{10 \text{ K}}{(0.8 \text{ K} \parallel 3 \text{ K}) + \frac{1}{3.67 \text{ m}}} \times 3.67 \text{ m} [10 \text{ K} \parallel 20 \text{ K} \parallel (3 \text{ K} + 0.8 \text{ K})]$$

$$= 11.06 \times 8.88 = 98.26 \text{ V/V}$$

(b)



$$\beta = \frac{V_f}{V_0} = \frac{R_1}{R_1 + R_2} = \frac{0.8}{0.8 + 3} = 0.21 \text{ V/V}$$

$$(c) A_f = \frac{A}{1 + \beta A} = \frac{98.26}{1 + 0.21 \times 98.26} = 4.54 \text{ V/V}$$

$$(d) R_{open} = R_{02} \parallel R_{04} \parallel (R_1 + R_2) \\ = 20 \text{ K} \parallel 10 \text{ K} \parallel (0.8 \text{ K} + 3 \text{ K}) \\ = 2.42 \text{ K}\Omega$$

$$R_{closed} = \frac{R_{open}}{1 + \beta A} = \frac{2.42 \text{ K}}{1 + 0.21 \times 98.26} = \underline{\underline{111.86 \Omega}}$$

Question 4 :

M_7 and M_8 will be operating in triode region

$$I_{D7} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[2(V_{GS} - V_{th}) V_{DS} - V_{DS}^2 \right]$$

$$0.5 \times 10^{-3} = \frac{1}{2} \times 135 \times 10^{-6} \times \frac{W}{L} \times \left[2(1.5 - 0.7) 0.1 - (0.1)^2 \right]$$

$$(W/L)_{7,8} = 49.38$$

$$I_{D5} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV5})^2$$

$$0.5 \times 10^{-3} = \frac{1}{2} \times 135 \times 10^{-6} \times 49.38 \times (V_{OV5})^2$$

$$V_{OV5} = 0.387 \text{ V}$$

$$\therefore V_{GS5} = 1.087 \text{ V}$$

$$\therefore V_{B1} = V_P + V_{GS5} = 0.1 + 1.087 = 1.187 \text{ V}$$

$$V_{out, min} = V_P + V_{OV5} + V_{OV1} = 0.1 + 0.387 + 0.387 = 0.874 \text{ V}$$

$$I_{D9} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{OV9})^2$$

$$0.5 \times 10^{-3} = \frac{1}{2} \times 40 \times 10^{-6} \times 49.38 \times (V_{OV9})^2$$

$$V_{OV9} = 0.71 \text{ V}$$

$$V_{out, max} = V_{DD} - V_{OV11} - V_{OV9}$$

$$= 3 - 0.71 - 0.71 = 1.58 \text{ V}$$

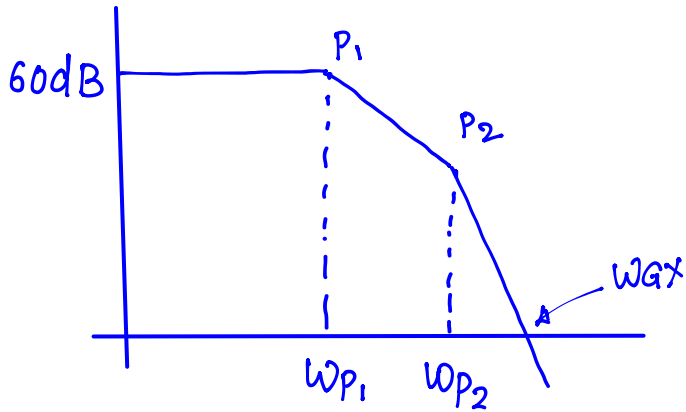
$$\text{Output voltage swing} = 2[V_{out, max} - V_{out, min}]$$

$$= 2[1.58 - 0.874]$$

$$= \underline{\underline{1.412 \text{ V}}}$$

Question 5 :

$$(a) \beta = 1 \quad \therefore \text{loop gain} = \beta A = 1 \cdot \frac{1000}{\left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)}$$



$$P_1 = (\omega_{p1}, 60 \text{ dB})$$

$$P_2 = (\omega_{p2}, x)$$

$$\frac{60 - x}{\log\left(\frac{1}{2}\right)} = -20$$

$$x = 53.98 \text{ dB}$$

$$\frac{53.98 \text{ dB} - 0 \text{ dB}}{\log\left(\frac{\omega_{p2}}{\omega_{GB}}\right)} = -40$$

$$\therefore \omega_{GB} = 22.36 \omega_{p2}$$

$$= 44.72 \text{ MHz}$$

$$\therefore \angle \beta A(j\omega) = -\tan^{-1}\left(\frac{44.72}{1}\right) - \tan^{-1}\left(\frac{44.72}{2}\right)$$

$$= -88.72 - 87.44 = -176.16$$

$$\therefore \text{PM} = 180 - 176.16 = \underline{\underline{3.84^\circ}}$$

$$P_1 = (\omega_{p1}, 60 \text{ dB})$$

$$P_2 = (\omega_{p2}, x)$$

$$x = 47.96 \text{ dB}$$

$$\frac{47.96}{\log\left(\frac{\omega_{p2}}{\omega_{GB}}\right)} = -40$$

$$\omega_{GB} = 63.25 \text{ MHz}$$

$$\angle \beta A(j\omega) = -\tan^{-1}\left(\frac{63.25}{1}\right) - \tan^{-1}\left(\frac{63.25}{4}\right)$$

$$= -89.09 - 86.38 = -175.47$$

$$\therefore \text{PM} = \underline{\underline{4.53^\circ}}$$

$$(b) \quad \beta = 1$$

$$|A_f(j\omega)| = \frac{|A(j\omega)|}{|1 + |\beta A(j\omega)| e^{j\angle \beta A(j\omega)}|}$$

At $\omega = \omega_{ax}$

$$|A_f(j\omega_{ax})| = \left| \frac{1}{1 + 1 \cdot e^{j0}} \right| = 1.5$$

$$\frac{1}{|1 + \cos 0 + j \sin 0|} = 1.5$$

$$\frac{1}{\sqrt{(1 + \cos 0)^2 + \sin^2 0}} = 1.5$$

$$\frac{1}{\sqrt{1 + 2\cos 0 + 1}} = 1.5$$

$$\frac{1}{2 + 2\cos 0} = (1.5)^2$$

$$1 = 4.5 + 4.5 \cos 0$$

$$0 = -141.1$$

$$\therefore \text{PM} = \underline{\underline{38.94^\circ}}$$